Rady | UC San Diego School of Management

MGT492R: Financial Risk Management

Rady School of Management 2025-26, UCSD

In person lectures: April 12th, May 10th, and June 7th, 8:00am-12:00pm
Location: 4N128
Remote asynchronous lectures: Canvas

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OFFICE HOURS: TBD (or ask for a Zoom link)
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COURSE DESCRIPTION

Risk management involves the application of modern financial techniques to assess and mitigate various types of risk using financial strategies and instruments. This course introduces the fundamental principles of risk management through lectures, case studies, assigned readings, and timely materials from the financial press.

We begin by examining whether, when, and how risk management enhances firm value. We then analyze key risk measures—such as Value-at-Risk (VaR) and Cash-Flow-at-Risk (CaR)—and their roles in project selection, performance evaluation, and managerial compensation in both financial and non-financial firms. The course next turns to strategies for managing financial risk exposures using derivatives (including options, futures, forwards, swaps, and credit default swaps), with applications to exchange rate risk, commodity price risk, interest rate risk, credit risk, climate risk, and operational risk. We also explore how strategic risk management can unlock the value of embedded real options. In addition, the course covers Basel II and other relevant banking regulations, along with real-world cases in which effective risk management is central to business success.

The course has a strong quantitative component. Students will work with financial data and implement risk management techniques, with a clear emphasis on practical application. Case studies play a central role in reinforcing concepts and addressing real-world business challenges.

Risk management is a rapidly expanding field. Those interested may consider joining the Global Association of Risk Professionals (GARP), which offers the Financial Risk Manager (FRM) accreditation—similar to the CPA for accountants and CFA for financial analysts. The course will incorporate select readings from GARP-recommended materials.

A working knowledge of basic statistical concepts—including means, variances, covariances, and linear regression—as well as proficiency in Excel, is assumed.

Prerequisites: Core Finance

SCHEDULE

Class will meet three times during the quarter, April 12th, May, 10th, and June 7th, from 8:00 to 12:00 with a coffee break. In addition, asynchronous lectures will be available for students on Canvas.

A recording of the class will be available on Canvas after the lecture.

Office hours will be held in person and/or via Zoom. If you cannot come in during office hours, please e-mail me for an appointment. I welcome questions via email.

CANVAS

You should already be familiar with Canvas. If not, please take the time to familiarize yourself with it. All class materials—including lecture slides, asynchronous videos, assignments, quizzes, additional resources—will be posted in a dedicated module.

COURSE MATERIAL

The lecture notes will be available on Canvas, posted before the lecture.

There is one required book as well as a course reader.

- Book: Philippe Jorion, Value at Risk, 3rd Edition, McGraw-Hill, 2006.
- Course Reader: Available on Canvas (Study.net)
- Optional Text: Options, Futures And Other Derivatives: Hull, John, 6th edition.

We will cover a few case studies from the course reader and/or additional reading material each session. Please keep up with the readings. If you have trouble accessing the material from the course reader, please let me know as soon as possible.

ASSIGNMENTS AND GRADING

There will be three case studies, which you will complete in groups. These case studies will be based on class discussions. Please format your responses to the case studies in a memo style, limiting the typed portion to a maximum of two pages. This page limit applies only to the written content. Attach all supporting materials (e.g., graphs, tables, pictures) at the end of the document and reference them appropriately within the memo's main body.

We will review and discuss the case study answers at the beginning of the following class. Be prepared to actively participate in these discussions.

There will be no midterm exams. However, the Final Exam will follow a case study format. You will have one week to turn it in. The exam must be completed individually, with no collaboration of any kind.

The grading policy is as follows:

1. Case Studies (40%). Group Assignment. One written assignment per group.

2. Quizzes (10%). Individual Assignment. Canvas, available after class.

3. Class Participation (10%). Attendance and class participation, contributing to a productive learning environment.

4. Final Examination (40%). Individual Assignment. The Final Exam will be a take-home case study, administered during Finals week, June 7th-June 12th.

To prevent one bad performance from affecting your grade, I will drop the lowest case study grade.

DETAILED SYLLABUS

We **may deviate** from the schedule depending on the pace the class feels comfortable with.

April 12th

Intro to Risk-Management; Market Risk; Why manage risk? Hedging; Diversification; Value-at-Risk (VAR); Cashflow-at-Risk (CAR). VAR of a portfolio. Expected Shortfall. Accuracy of VARs. Review of Derivatives: Forwards, Futures, Options

Reading:

Jorion, Ch 4, 5, 6, 7
Rethinking Risk Management, Stulz, Journal of Applied Corporate Finance
A Comparables Approach to Measuring Cashflow-at-Risk for Non-Financial Firms
An Overview of Value-at-Risk (Duffie and Pan), The Journal of Derivatives
How Accurate are Value-at-Risk Models at Commercial Banks (Berkowitz and O'Brien)
A Framework for Risk Management, Froot, Scharfstein, and Stein, Harvard Business Review
Options. Their use in risk-management.
Fact and Fantasy in the Use of Options, Fischer Black, Financial Analyst Journal
Leland, O'Brien, Rubinstein Associates Incorporated: Portfolio Insurance, HBS 9-294-061

Case Study 1:

Why Manage Risk? HBS 9-294-107
Due: April 26th, 10 pm

Asynchronous:

- **Value at Risk (in more detail) – available from start**
- **Case Study 1: Discussion and Solutions – available after April 26th, 10:01pm**

May 10th

Hedging; Optimal hedge ratio; Futures, Forwards, and Options. Hedging with Options, Dynamic Hedging; Measuring and Managing Exchange Rate Risk. Measuring and Managing Commodity Price Risk

Reading: American Barrick Resources Corporation: Managing Gold Price Risk; HBS 9-293-128.
Note on Crude Oil and Crude Oil Derivative Markets.
Replicating Options with Positions in Stock and Cash, Rubinstein and Leland, Financial Analyst Journal
Identifying, Measuring, and Hedging Currency Risk at Merck, Lewent, Kearney, Journal of Applied Corporate Finance
The New Risk Management: The Good, the Bad, and the Ugly (Dybvig and Marshall), Federal Reserve Bank of St. Louis.

Case Study 2: Credit General, SA, HBS 9-296-011
Due: May 24th, 10 pm

Asynchronous:

- **Derivatives: Options and Futures – available from start**
- **Review Pricing of Forward /Futures – available from start**
- **Case Study 2: Discussion and Solutions – available after May 24th, 10:01pm**

June 7th Fixed Income; Interest Rate Risk; Immunization; Hedging Interest Rate Risk with Futures and Forwards; Credit/Default Risk; Hedging credit risk with credit default swaps (CDS). Regulatory Risk.

Reading: Jorion, Ch 19, 20, 21, 22
Amendment to the Capital Accord to Incorporate Market Risks, Basle Committee on Banking Supervision 1996.
If time allows: Operational Risk Management, Basle Committee on Banking Supervision 1998.
An Overview of Credit Derivatives, HBS 9-297-086
Enron Corp-Credit Sensitive Notes, 9-297-099

Case Study 3: Orange County, Jorion's website.
Due: June 10th, 10 pm.

Asynchronous:

- **Credit Risk – available from start**
- **Case Study 3 – available after June 10th, 10.01pm**

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Final Exam: **Format: Case Study**
Distributed: June 10th, 10.01pm (on Canvas)
Due: June 13th, 10 pm (turn in on Canvas)

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ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind. The complete UCSD Policy on Integrity of Scholarship can be viewed at:

http://wwwsenate.ucsd.edu/manual/appendices/app2.htm#AP14

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member. The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.**

Please visit the OSD website for further information or contact the Office for Students with Disabilities at(858) 534-4382 or osd@ucsd.edu.